Training a small image model

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Teach exactly what was fitted and which data informed each choice.

Supply complete public development data and private teacher final references;keep actual freeze/review status and synthetic coverage limits explicit.

Keep one habit: Assess the independent checkpoint calculation and require a full per-case frozen evaluation before broader claims.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Trace a complete image model with fixed features and one trained weight

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Moving the same values between left and right can preserve the maximum and mean. A summary of all four positions may discard the distinction the task needs.

Opening retrieval

Without opening the old lesson, explain one key idea from Pooling and spatial information. Give an example and a mistake that the idea helps you avoid.

Look for: Compute max and mean pooling under an explicit window rule

Reference example, not the only acceptable wording: Teacher explanation: Values are 1, 3, 2 and 7. Sum is 13, maximum 7 and mean 3.25. Each pooled **output** has shape 2 by 2.

Without opening the old lesson, explain one key idea from Image tensors and channels. Give an example and a mistake that the idea helps you avoid.

Look for: Read a pixel using declared row column and channel axes

Reference example, not the only acceptable wording: Teacher explanation: The RGB pixel is [0, 0, 255]; its blue value is 255. Shape is 2 by 2 by 3, with four pixels and twelve channel values.

Without opening the old lesson, explain one key idea from Initialisation and reproducibility. Give an example and a mistake that the idea helps you avoid.

Look for: Trace how starting values affect a fixed **training** budget

Reference example, not the only acceptable wording: Teacher explanation: States are -2, 0 and 1. Final **training** MSE is 2.5. Each step halves the signed distance from 2.

Pictured end goal

By the end, I can explain **training** a small image model using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Pooling and spatial information

Image tensors and channels

Initialisation and reproducibility

For the compact alternative schedule, use two 45-minute sessions. The model deliberately trains only one **output** scalar; do not present the fixed kernels as learned.

Keep the final cases below teacher-only until students record the selected **checkpoint** and threshold. Supply each final image first for **prediction**, then reveal its label.

No image download, account or external **training** library is needed.

Outcomes

Trace a complete image model with fixed features and one trained weight

Select a **checkpoint** using a declared validation rule

Evaluate a frozen model against a **baseline** and limit the conclusion

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Materials and actual final-release gate

Prepare the six public A–Fpixel cards and the complete student program. Keep the G–Ireference fixture and evaluator in teacher materials until the selected checkpoint and full procedure have actually been recorded. Provide a paper trace if code cannot be run, and distinguish that trace from observed execution. The supplied reference scores do not stand in for a learner's own project results or actual review status.

MAIN LESSON · SECTION 2

Teach fixed features,training,selection and final release in order

> Keep actual final cases private during development modelling.

Main model,complete register/checkpoints and guided practice key

Use fictional 2 by 2 binary images, stored as two rows. Predict class 1 when the left column sum exceeds the right column sum; otherwise predict class 0. Equal sums belong to class 0.

This known arithmetic task makes every step checkable; it does not establish real-world image recognition.

Apply the fixed kernel [[1, -1], [1, -1]] to the entire image to obtain d. Its negative gives -d. ReLU produces h1 = max(0, d) and h2 = max(0, -d). The **output** score is a(h1 - h2) = ad.

Only the shared scalar a is trained; both kernels stay frozen and there is no bias or pooling.

For squared-error **training**, map class 1 to target t = 1 and class 0 to t = -1. Predict class 1 only when score > 0; otherwise class 0. The score is not a probability.

Keep the teacher-held final cases unseen until the **checkpoint** is selected and recorded.

Start a = 0. Use all four **training** examples in each update, ordinary mean squared error and learning rate 0.1. Save **checkpoint**s at updates 0, 1 and 2. The **training** loss is 2.5a squared - 3a + 1, with gradient 5a - 3.

Subtract learning rate times gradient; this gives a = 0.3 then a = 0.45.

The code assumes valid 2 by 2 numeric images and nonempty labelled lists. It prints update, weight, **training** MSE and validation MSE. Validation examples are scored but never used in the gradient.

A single-channel filter spanning the whole image produces one value here, so no further spatial pooling is needed.

Model component	Complete definition / training status
Input	One valid 2×2 binary image, rows then columns
Frozen kernels	K[[1, −1], [1, −1]]computes d; its negative computes−d
Frozen activation	h=[ReLU(d), ReLU(−d)], so h1−h2=d
Trainable output	Score a(h1−h2)=ad; coefficients[a, −a]are tied to one scalar, not two independent weights
Loss and targets	Ordinary MSE, class 1 target+1, class 0 target−1
Prediction	Class1 when score>0; otherwise 0, including equality
Training protocol	Four training rows, all per update, starta 0, rate.1, two updates; no bias or pooling
Selection and baseline	Lowest observed validation MSE, earlier exact ties; training-majority class baseline chooses 0on a tie

Updates completed · a · Gradient for next update · Training MSE · Validation MSE · Training class correct/4 · Validation class correct/2

> **Updates completed:** 0
>
> **a:** 0
>
> **Gradient for next update:** -3
>
> **Training MSE:** 1
>
> **Validation MSE:** 1
>
> **Training class correct/4:** 2
>
> **Validation class correct/2:** 1

> **Updates completed:** 1

> **a:** 0.3
>
> **Gradient for next update:** -1.5
>
> **Training MSE:** 0.325
>
> **Validation MSE:** 0.49
>
> **Training class correct/4:** 4
>
> **Validation class correct/2:** 2

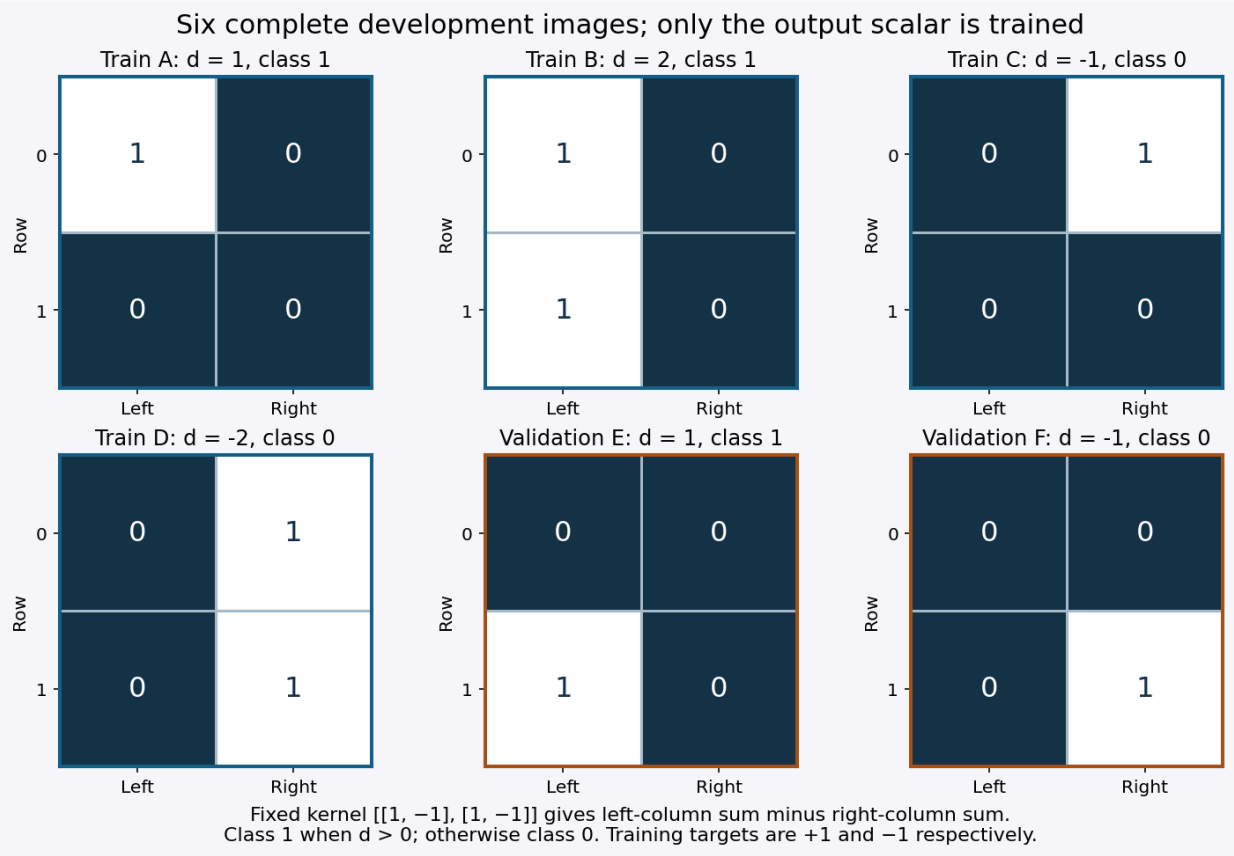
> **Updates completed:** 2
>
> **a:** 0.45
>
> **Gradient for next update:** Stop at declared budget
>
> **Training MSE:** 0.15625
>
> **Validation MSE:** 0.3025
>
> **Training class correct/4:** 4
>
> **Validation class correct/2:** 2

The public program supplies all six A–Fdevelopment images, performs the two training updates, selects checkpoint 2(a≈.45)using validation MSE, and audits the frozen class 0 baseline. The two positive checkpoints have the same classification accuracy but different squared losses. Scores are not probabilities. Display rounding makes floating-point output readable; decisions use unrounded scores.

These constructed arrays and reused differences do not establish independent real-object coverage or broad image recognition. The three final cases and references remain in teacher materials until the actual freeze record is checked.

Current guided key only:at a.3, E/Fscores±.3 and at a.45 scores±.45; both classify E1/F0 correctly. Validation MSEchanges.49→.3025, so the declared rule selects update 2. Train classes 2/2 tie gives baseline 0. Keep the separate final-case reference folded until the learner’s actual freeze record is checked.

All public development images



Separate pictured one-pair experiment

Train one tiny image score with a fixed budget and a separately reserved check.

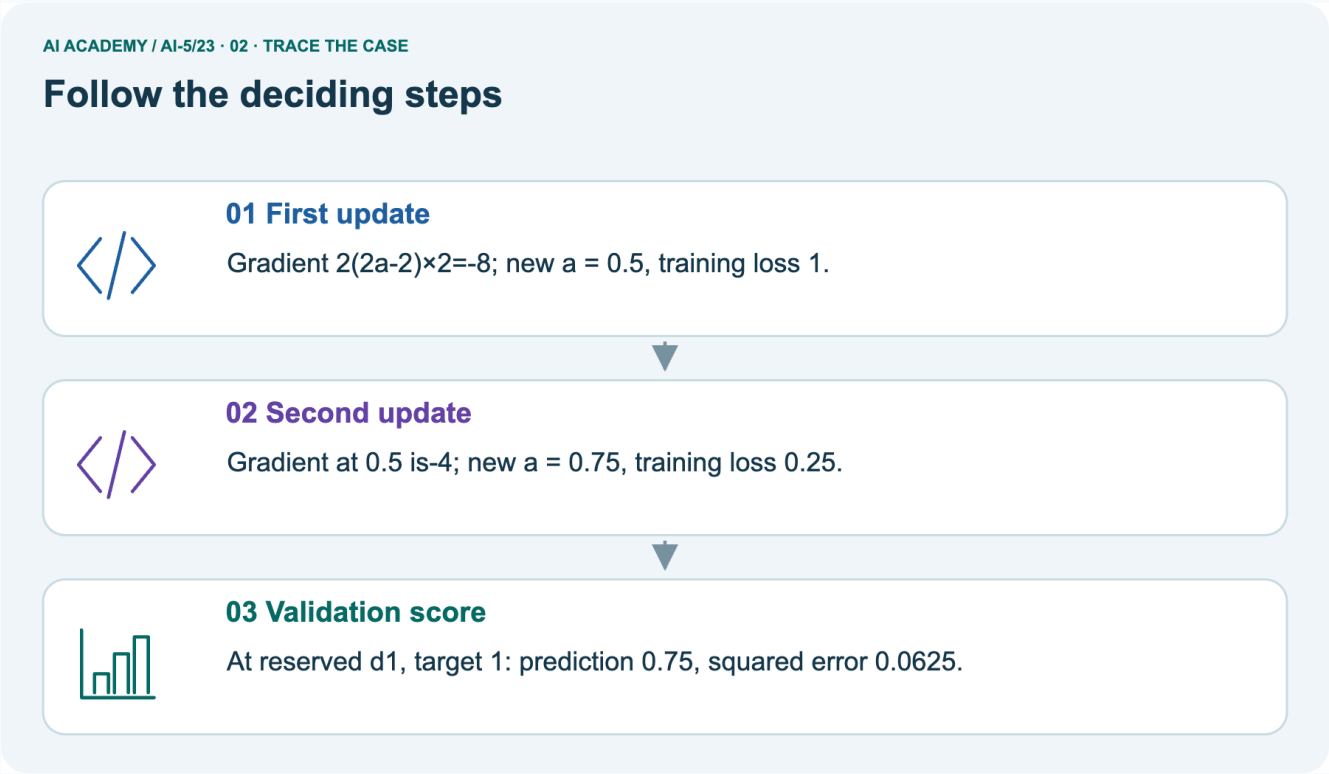
Toy brightness difference d=left−right; score=a×d. **Training** pair d=2, target 2; start a=0; squared loss; learning rate 1/16; two updates.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture is a separate scalar brightness experiment:d2, target 2, starta 0, squared loss, rate 1/16. Two updates use gradients−8 then−4, giving a.5 then.75 and training losses 1 then.25. Its supplied reserved d1, target 1 pair has score.75, error.0625. Four updates instead reach.875 then.9375, training loss.015625.

This different target/rate does not provide the main image model's final cases or a comparison of independent performance after choosing the budget.

Trace the training-budget change without a generalization claim



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. First update	Gradient 2(2a-2)×2=-8; new a = 0.5, training loss 1.
2. Second update	Gradient at 0.5 is-4; new a = 0.75, training loss 0.25.
3. Validation score	At reserved d1, target 1: prediction 0.75, squared error 0.0625.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

The image feature is a brightness difference, and **training** adjusts one **output** weight multiplying it. Each gradient is computed using the current **prediction** before the next update.

In this example, the rate halves the distance to the **training** optimum each time. More updates reduce this **training** loss, but choosing a budget still requires separate validation **evidence**.

This scalar brightness model is a transparent toy image task, not a claim that a full image network was run.

Change: Double the two-update budget while keeping the same **training** rule.

Prediction: Next a values 0.875 and 0.9375; **training** loss falls to 0.015625.

Reason: Each update halves the distance to the **training** optimum 1 in this toy system.

▼ Teacher-only final-case reference after freeze

Guided: at a = 0.3, E and F score 0.3 and -0.3; at a = 0.45 they score 0.45 and -0.45. Both **checkpoint**s predict classes 1 and 0 correctly, while squared residuals change from 0.49 to 0.3025.

Final cases: G rows [1, 1], [1, 0], class 1; H rows [1, 1], [0, 1], class 0; I rows [1, 0], [0, 1], class 0. Differences are 1, -1, 0; scores 0.45, -0.45, 0; **prediction**s 1, 0, 0. Model accuracy is 3 of 3, **baseline** 2 of 3.

Signed-target squared errors are 0.3025, 0.3025, 1, so final MSE is 0.535. The tie case is classified correctly despite nonzero regression error.

Exit: only a was learned; using final results to choose a model contaminates final evaluation. Preserve setup, **prediction**s, **baseline** and limits.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Frozen feature	A feature computation whose parameters are kept unchanged during training
Checkpoint	Saved model parameters at a particular training stage
Selection metric	The quantity used under a declared rule to choose a model
Baseline	A simple comparison method evaluated on the same cases

Compare the saved checkpoints

UPDATE AND WEIGHT	TRAINING MSE	VALIDATION MSE
0 with a 0	1	1
1 with a 0.3	0.325	0.49
2 with a 0.45	0.15625	0.3025

Before training, declare selection by lowest validation MSE among these three checkpoints, with the earliest checkpoint winning an exact tie. Checkpoint 2 is selected. Training loss alone does not define the selection rule.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Compare the saved checkpoints

UPDATE AND WEIGHT	TRAINING MSE	VALIDATION MSE
0 with a 0	1	1
1 with a 0.3	0.325	0.49
2 with a 0.45	0.15625	0.3025

Before training, declare selection by lowest validation MSE among these three checkpoints, with the earliest checkpoint winning an exact tie. Checkpoint 2 is selected. Training loss alone does not define the selection rule.

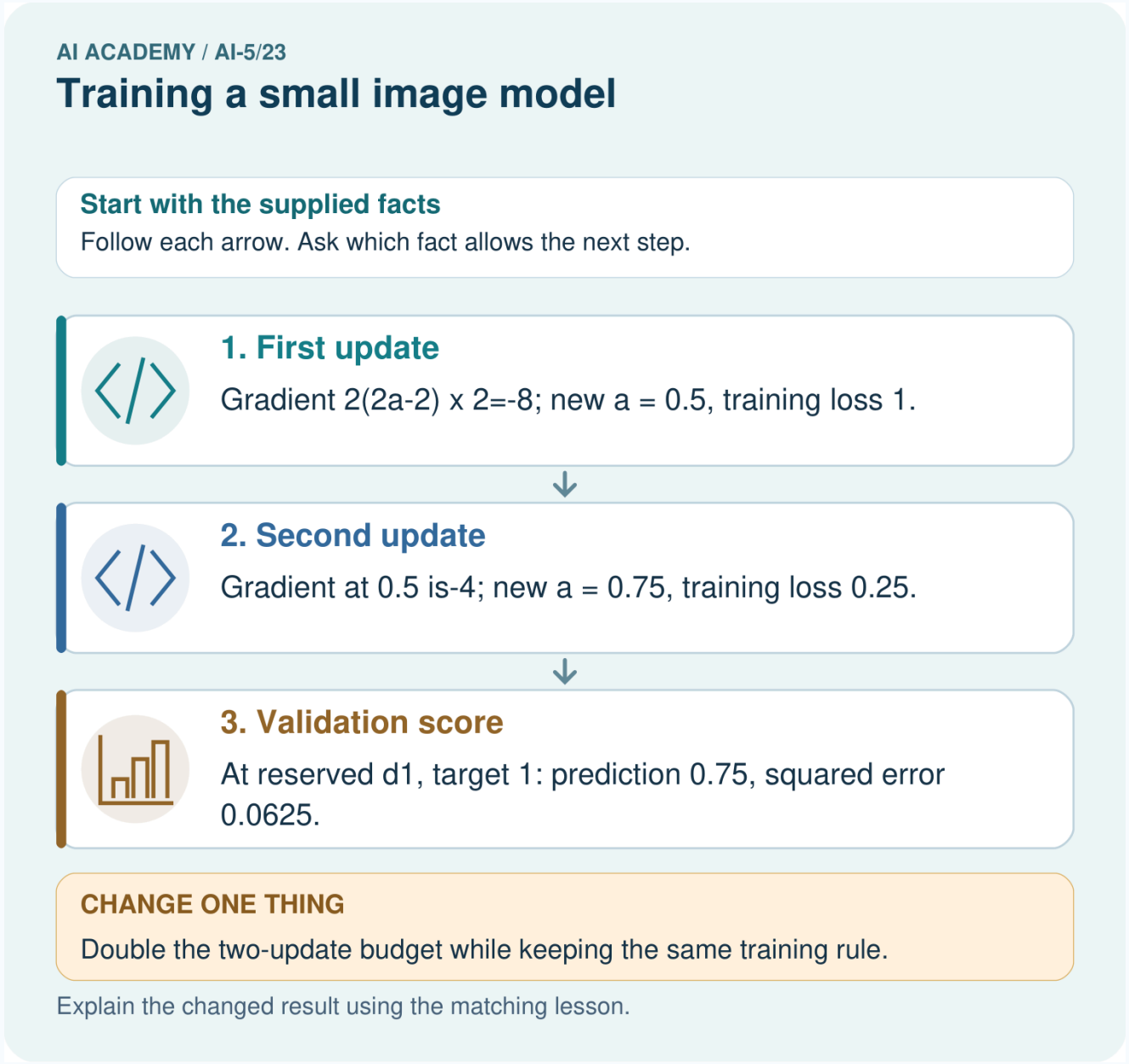
At a = 0.3 and a = 0.45, compute the validation scores for E and F and their predicted classes. Why can validation accuracy stay the same while validation MSE changes?

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Training a small image model

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Teacher-only complete final evaluator and checked result

Teacher final reference only:after checking the actual saved checkpoint 2/a≈.45 system, the supplied final G/H/Icases yield scores.45/−.45/0, classes 1/0/0, model 3/3 versus fixed-class 0 baseline2/3. Signed-target MSEis.535; the equal-column case is classified correctly despite squared error 1.

The executable reproduces this fictional reference calculation; it does not assert a completed classroom approval or broad image-recognition capability. Keep the full ledger and do not retune on these cases while calling a revised result untouched.

```
def feature(image):
    return image[0][0] + image[1][0] - image[0][1] - image[1][1]

train = [([[1, 0], [0, 0]], 1),
         ([[1, 0], [1, 0]], 1),
         ([[0, 1], [0, 0]], -1),
         ([[0, 1], [0, 1]], -1)]
valid = [([[0, 0], [1, 0]], 1),
         ([[0, 0], [0, 1]], -1)]

def mse(a, rows):
    return sum((a * feature(x) - t)**2 for x, t in rows) / len(rows)

a, checkpoints = 0.0, [0.0]
for _ in range(2):
    gradient = sum(2 * (a * feature(x) - t) * feature(x)
                   for x, t in train) / len(train)
    a -= 0.1 * gradient
    checkpoints.append(a)
for step, value in enumerate(checkpoints):
```

```
    print(step, round(value, 6), round(mse(value, train), 6),
          round(mse(value, valid), 6))
# Select using validation MSE only; exact ties keep the earlier checkpoint.
best_index = min(range(len(checkpoints)), key=lambda i: (mse(checkpoints[i],valid),i))
selected = checkpoints[best_index]
positive = sum(t == 1 for _,t in train)
negative = sum(t == -1 for _,t in train)
baseline_class = 1 if positive > negative else 0  # Declared class-0 tie rule.

def prediction(weight, image):
    d = feature(image)
    hidden = [max(0,d),max(0,-d)]
    score = weight*(hidden[0]-hidden[1])
    return d,hidden,score,int(score > 0)

def audit(weight, rows, ids, baseline):
    if len(rows) != len(ids) or not rows:
        raise ValueError('Need one ID per nonempty supplied case list')
    ledger = []
    for name,(image,target) in zip(ids,rows):
        d,h,score,pred = prediction(weight,image)
        ledger.append({'id':name,'d':d,'hidden':h,'score':score,
                       'class':int(target==1),'prediction':pred,'baseline':baseline})
    return ledger

print('selected checkpoint / weight', best_index,round(selected,6))
print('training class counts 1/0', positive,negative,'frozen baseline',baseline_class)
for split,rows,ids in [('training',train,['A','B','C','D']),('validation',valid,
['E','F'])]:
    ledger = audit(selected,rows,ids,baseline_class)
    for row in ledger:
        print(split,{**row,'score':round(row['score'],6)})  # Display rounding only.
    correct = sum(r['prediction']==r['class'] for r in ledger)
    base_correct = sum(r['baseline']==r['class'] for r in ledger)
    print(split,'candidate / baseline correct / total',correct,base_correct,len(ledger))
# This public boundary probe is not one of the teacher-held final cases.
empty_pattern = [[0,0],[0,0]]
print('public equal-column-sum probe
d/hidden/score/class',prediction(selected,empty_pattern))
print('Final cases absent. Save selected parameters and protocol before teacher release.')
# Fixed kernels, no bias/pooling; only a was fitted. Finite valid binary images assumed.

# TEACHER ONLY: inspect the actual saved freeze record before releasing these cases.
assert best_index == 2 and abs(selected-.45) < 1e-12
final = [([[1,1],[1,0]],1), ([[1,1],[0,1]],-1), ([[1,0],[0,1]],-1)]
final_ledger = audit(selected,final,['G','H','I'],baseline_class)
for row in final_ledger:
    print('TEACHER frozen final',{**row,'score':round(row['score'],6)})
```

```
print('TEACHER candidate / baseline correct / total',
      sum(r['prediction']==r['class'] for r in final_ledger),
      sum(r['baseline']==r['class'] for r in final_ledger),len(final_ledger))
print('TEACHER candidate signed-target MSE',round(mse(selected,final),6))
# This reference run does not assert a real classroom approval or support retuning on
final errors.
```

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Starting at a = 0, use gradient 5a - 3 and rate 0.1 for two updates. Give each gradient, new weight and **training** MSE.

Teacher check: Teacher explanation: First gradient -3 gives a = 0.3 and MSE 0.325. Second gradient -1.5 gives a = 0.45 and MSE 0.15625. Each gradient is computed at the current weight before the update.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Training** a small image model. Proposed explanation: “All parameters in a trained model must have been learned”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: The declared experiment freezes the feature kernels and adjusts only a.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Does lower **training** loss alone establish that four updates generalise better? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

No. Record the extra budget and score suitable validation **evidence**; use an untouched final evaluation after selecting a schedule. Changed-case check: Next a values 0.875 and 0.9375; **training** loss falls to 0.015625.

Each update halves the distance to the **training** optimum 1 in this toy system.

Changed-case reasoning: Next a values 0.875 and 0.9375; **training** loss falls to 0.015625. Each update halves the distance to the **training** optimum 1 in this toy system.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide a ledger for image, difference, signed target, score, residual and class. Allow calculators and distinguish class 0 from signed target -1.

Extension: enumerate all sixteen binary 2 by 2 images under the frozen model and exact task rule; label this exhaustive toy-domain audit separately from the three-case held-out evaluation.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve the pooling collision and declare the binary image task, signed targets and separate splits.
7 to 18 minutes	Trace fixed kernels, ReLU features and the scalar head, then complete W1.
18 to 32 minutes	Compute both full-data updates and use W2 to check the loss arithmetic.
32 to 45 minutes	Run or trace the code and use W3 to select a saved checkpoint by validation MSE.
Session 2 start to 8 minutes	Complete the guided validation scores; record the frozen model and constant baseline.
8 to 20 minutes	Reveal final images G, H and I only after the record is complete; collect predictions before labels, then compare both methods.
20 to 35 minutes	Collect W4 independently, requiring validation calculations for every checkpoint.
35 to 45 minutes	Use W5 and the exit response to audit claims and administer the fresh retry.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main gradients−3 then−1.5 givea.3/.45, MSE.325/.15625; validation selects update 2. Only outputa fitted. Teacher final fixture is in the private reference section, not current student practice. Independentb*dtrainingd±1, t±1, start 0, gradient 2(b−1), rate.25, two updates→b0/.5/.75; validationd±2, t±1 givesMSE1/0/.25, selectupdate 1/b.5. Record separate split roles and preserve any final error before revision.

Diagnose the current misconceptions

All parameters in a trained model must have been learned: The declared experiment freezes the feature kernels and adjusts only a.

Perfect accuracy means zero MSE: Thresholded classes can all be correct while continuous scores differ from signed targets.

The last **checkpoint** must be best: Selection depends on the declared validation rule, not update order or **training** loss alone.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: A separate model scores bd. Its **training** examples have d = 1 and -1 with signed targets 1 and -1. Start b = 0; gradient is 2(b - 1), rate 0.25 and budget two updates.

Validation differences are 2 and -2 with targets 1 and -1. Calculate all three weights and their validation MSEs, then select by lowest validation MSE.

Teacher explanation: Weights are 0, 0.5 and 0.75. Validation scores are [0, 0], [1, -1] and [1.5, -1.5], so validation MSEs are 1, 0 and 0.25. Select update 1 with b = 0.5, even though a later update exists.

Training MSE falls from 1 to 0.25 to 0.0625.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Trace features and a prediction

For Train B, compute d, h1, h2 and the score at a = 0.45. Give the class **prediction** and squared error against its signed target.

Teacher explanation: d = 2, h1 = 2 and h2 = 0. Score is 0.9, **prediction** class 1 and squared error (0.9 - 1) squared = 0.01.

W2 Check both updates

Starting at a = 0, use gradient 5a - 3 and rate 0.1 for two updates. Give each gradient, new weight and **training** MSE.

Teacher explanation: First gradient -3 gives a = 0.3 and MSE 0.325. Second gradient -1.5 gives a = 0.45 and MSE 0.15625. Each gradient is computed at the current weight before the update.

W3 Apply the selection rule

Select among the three saved lesson **checkpoint**s using validation MSE, with earliest update breaking a tie. Is selecting by **training** accuracy the same declared procedure?

Teacher explanation: Select update 2 with a = 0.45 and validation MSE 0.3025.

Training accuracy is a different **selection metric**; both trained **checkpoint**s have perfect **training** accuracy, so it does not establish the same choice.

W5 Audit a final evaluation

After inspecting the final cases, a team changes the decision threshold and reports the best result as an unseen test score. Explain the problem and list two things the original report should preserve.

Teacher explanation: Final outcomes influenced the revised model, so that set is no longer unseen **evidence** for the revision.

Preserve the original frozen **checkpoint** and threshold, per-case **prediction**s and errors, **baseline** results, and split IDs. Evaluate the revision on new reserved cases.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: at a = 0.3, E and F score 0.3 and -0.3; at a = 0.45 they score 0.45 and -0.45. Both **checkpoint**s predict classes 1 and 0 correctly, while squared residuals change from 0.49 to 0.3025.

Final cases: G rows [1, 1], [1, 0], class 1; H rows [1, 1], [0, 1], class 0; I rows [1, 0], [0, 1], class 0. Differences are 1, -1, 0; scores 0.45, -0.45, 0; **prediction**s 1, 0, 0. Model accuracy is 3 of 3, **baseline** 2 of 3.

Signed-target squared errors are 0.3025, 0.3025, 1, so final MSE is 0.535. The tie case is classified correctly despite nonzero regression error.

Exit: only a was learned; using final results to choose a model contaminates final evaluation. Preserve setup, **prediction**s, **baseline** and limits.

W1 Trace features and a prediction

W2 Check both updates

W3 Apply the selection rule

W4 Choose a checkpoint independently

A separate model scores bd. Its **training** examples have d = 1 and -1 with signed targets 1 and -1. Start b = 0; gradient is 2(b - 1), rate 0.25 and budget two updates. Validation differences are 2 and -2 with targets 1 and -1.

Calculate all three weights and their validation MSEs, then select by lowest validation MSE.

Teacher explanation: Weights are 0, 0.5 and 0.75. Validation scores are [0, 0], [1, -1] and [1.5, -1.5], so validation MSEs are 1, 0 and 0.25. Select update 1 with b = 0.5, even though a later update exists.

Training MSE falls from 1 to 0.25 to 0.0625.

W5 Audit a final evaluation

Independent assessment and fresh retry

W4 requires weights 0, 0.5, 0.75 and validation MSEs 1, 0, 0.25; select update 1. Fresh retry: train on d = 2 and -2, signed targets 1 and -1; gradient 8b - 4, rate 0.0625, start 0 and two updates. Weights are 0, 0.25, 0.375.

Validate on d = 4 and -4 with targets 1 and -1: MSEs are 1, 0, 0.25. Select update 1 with b = 0.25.

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled **training** a small image model **evidence** sheet to Image-pattern Network Investigation. Show how you can trace a complete image model with fixed features and one trained weight.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide a ledger for image, difference, signed target, score, residual and class. Allow calculators and distinguish class 0 from signed target -1.

Extension: enumerate all sixteen binary 2 by 2 images under the frozen model and exact task rule; label this exhaustive toy-domain audit separately from the three-case held-out evaluation.

Week 24 examines augmentation and leakage, including how related image variants should remain in the same split.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/24, ai-5/26, ai-5/29. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Trace a complete image model with fixed features and one trained weight

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Starting at a = 0, use gradient 5a - 3 and rate 0.1 for two updates. Give each gradient, new weight and **training** MSE.

Reasoned practice answer: Teacher explanation: First gradient -3 gives a = 0.3 and MSE 0.325. Second gradient -1.5 gives a = 0.45 and MSE 0.15625. Each gradient is computed at the current weight before the update.

Check every learner before discussion: Use the worked case for **Training** a small image model. Proposed explanation: “All parameters in a trained model must have been learned”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: The declared experiment freezes the feature kernels and adjusts only a.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Double the two-update budget while keeping the same **training** rule. Require a **prediction** and a reason before comparing. Expected result: Next a values 0.875 and 0.9375; **training** loss falls to 0.015625.

Reason: Each update halves the distance to the **training** optimum 1 in this toy system.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project **checkpoint**: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

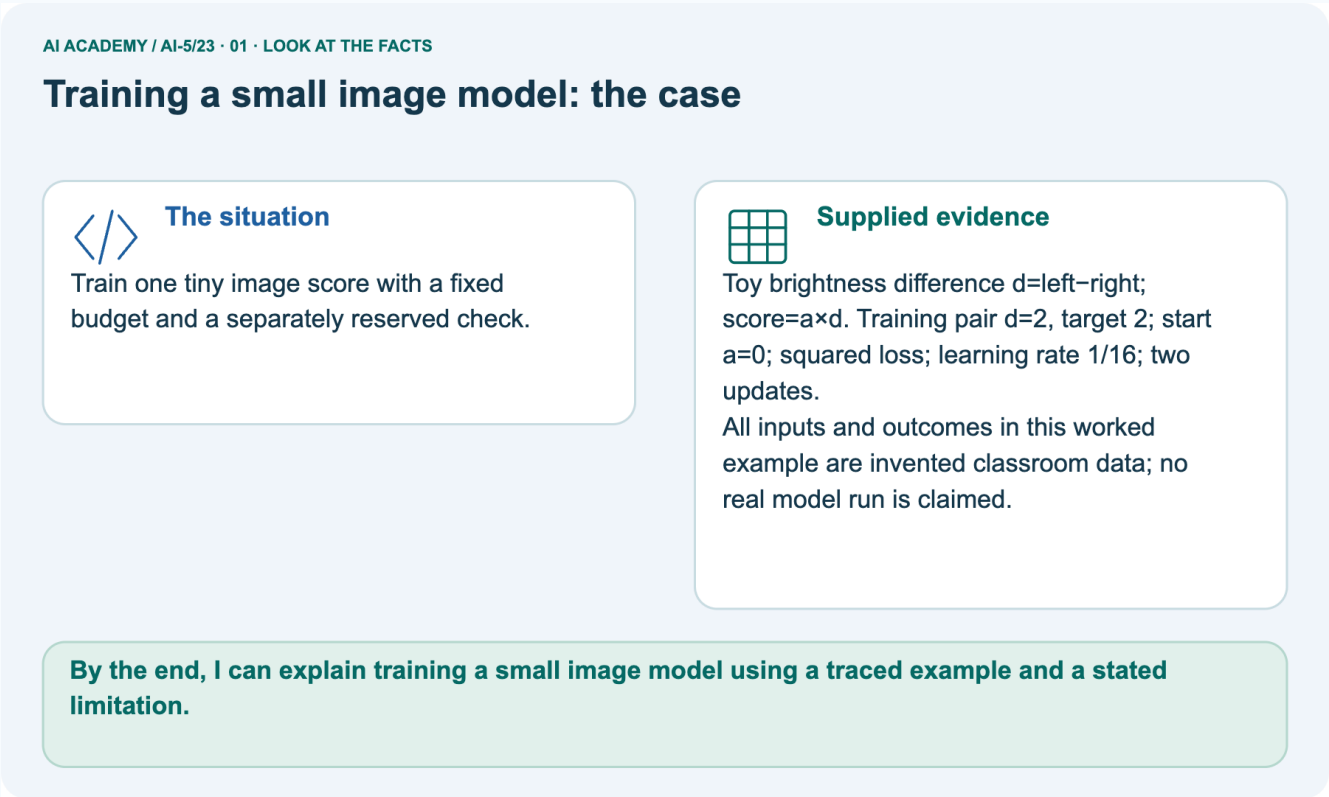
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Training a small image model: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

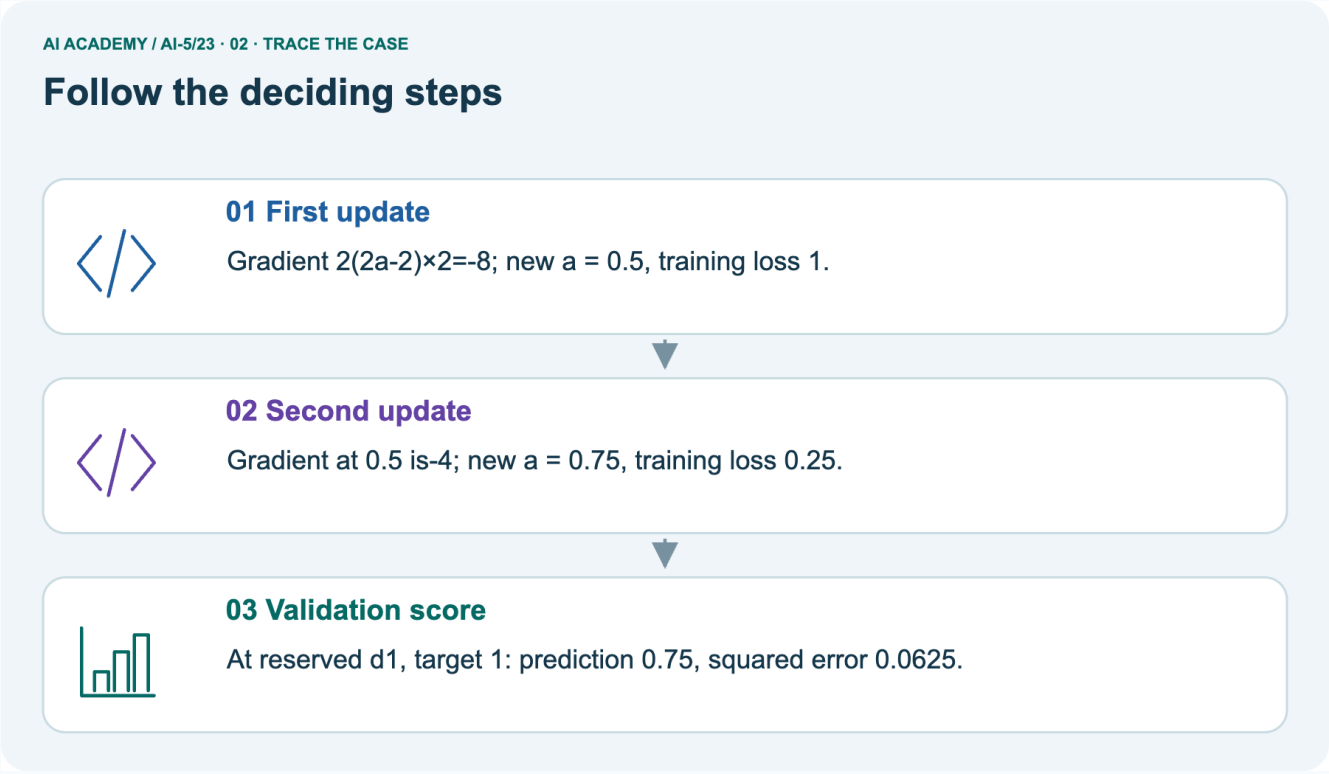
Train one tiny image score with a fixed budget and a separately reserved check. Supplied facts: Toy brightness difference d=left−right; score=a×d.

Training pair d=2, target 2; start a=0; squared loss; learning rate 1/16; two updates. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

First update: Gradient 2(2a-2)×2=-8; new a = 0.5, **training** loss 1. → Second update: Gradient at 0.5 is-4; new a = 0.75, **training** loss 0.25. → Validation score: At reserved d1, target 1: **prediction** 0.75, squared error 0.0625.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Compare the saved checkpoints

AI ACADEMY / AI-5/23 · 03 · INSPECT THE LESSON MODEL

Compare the saved checkpoints

Update and weight	Training MSE	Validation MSE
0 with a 0	1	1
1 with a 0.3	0.325	0.49
2 with a 0.45	0.15625	0.3025

Before training, declare selection by lowest validation MSE among these three checkpoints, with the earliest checkpoint winning an exact tie. Checkpoint 2 is selected. Training loss alone does not define the selection rule.

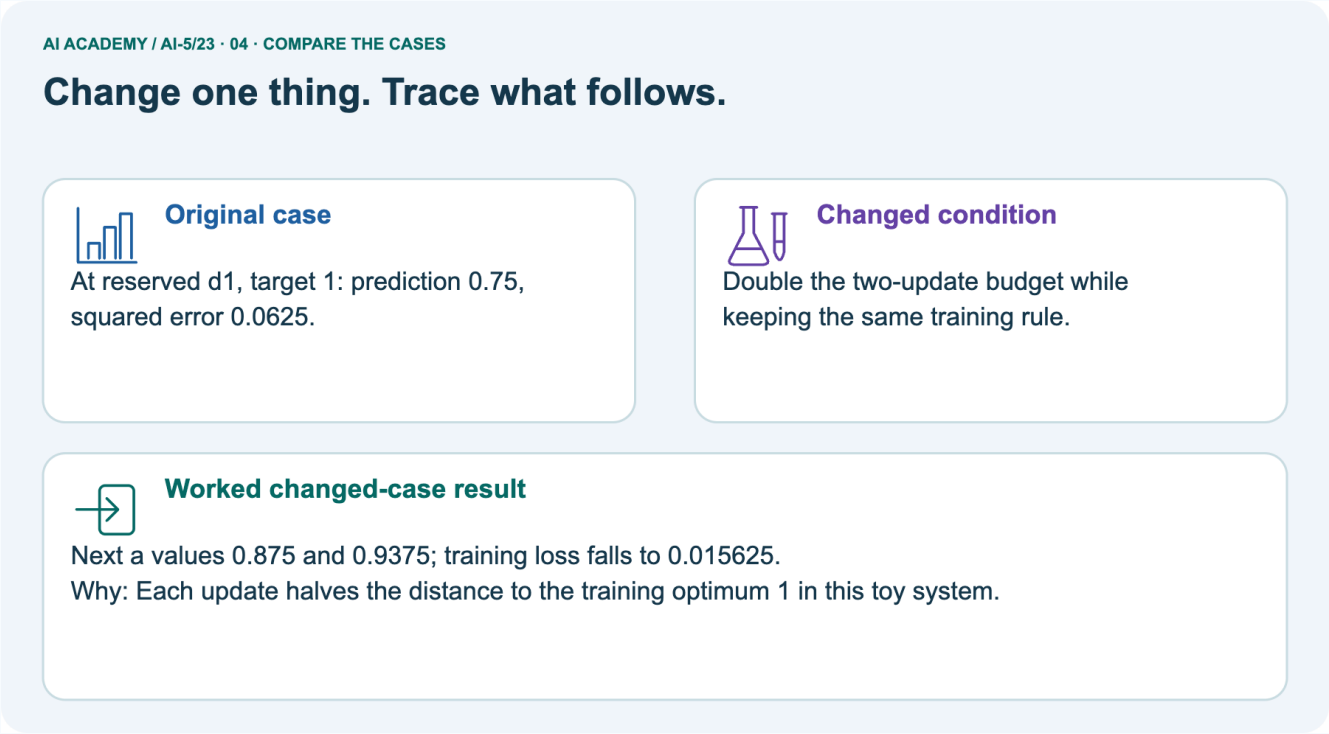
Follow the labelled picture with the matching explanation. Open the vector diagram

Compare the saved **checkpoint**s. 0 with a 0 | 1 | 1. 1 with a 0.3 | 0.325 | 0.49. 2 with a 0.45 | 0.15625 | 0.3025

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: At reserved d1, target 1: **prediction** 0.75, squared error 0.0625. Changed condition: Double the two-update budget while keeping the same **training** rule.

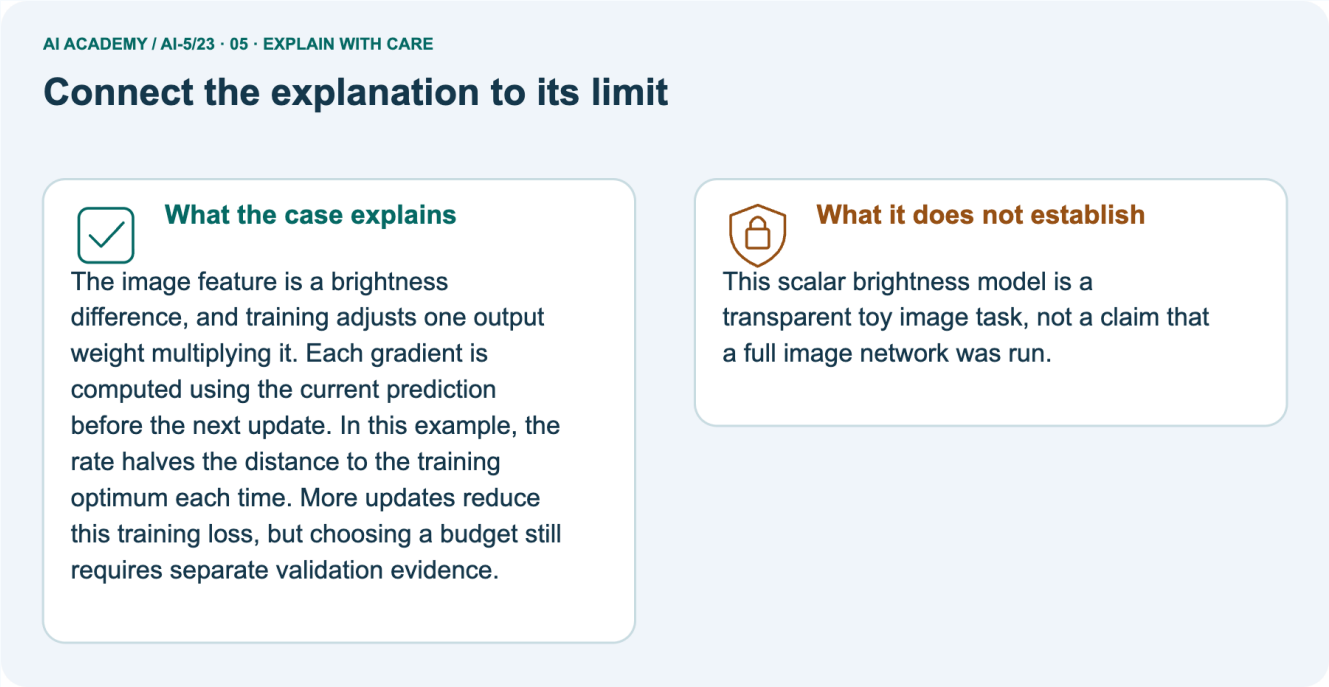
Expected result: Next a values 0.875 and 0.9375; **training** loss falls to 0.015625. Reason: Each update halves the distance to the **training** optimum 1 in this toy system.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Each update halves the distance to the **training** optimum 1 in this toy system.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The image feature is a brightness difference, and **training** adjusts one **output** weight multiplying it. Each gradient is computed using the current **prediction** before the next update.

In this example, the rate halves the distance to the **training** optimum each time. More updates reduce this **training** loss, but choosing a budget still requires separate validation **evidence**.

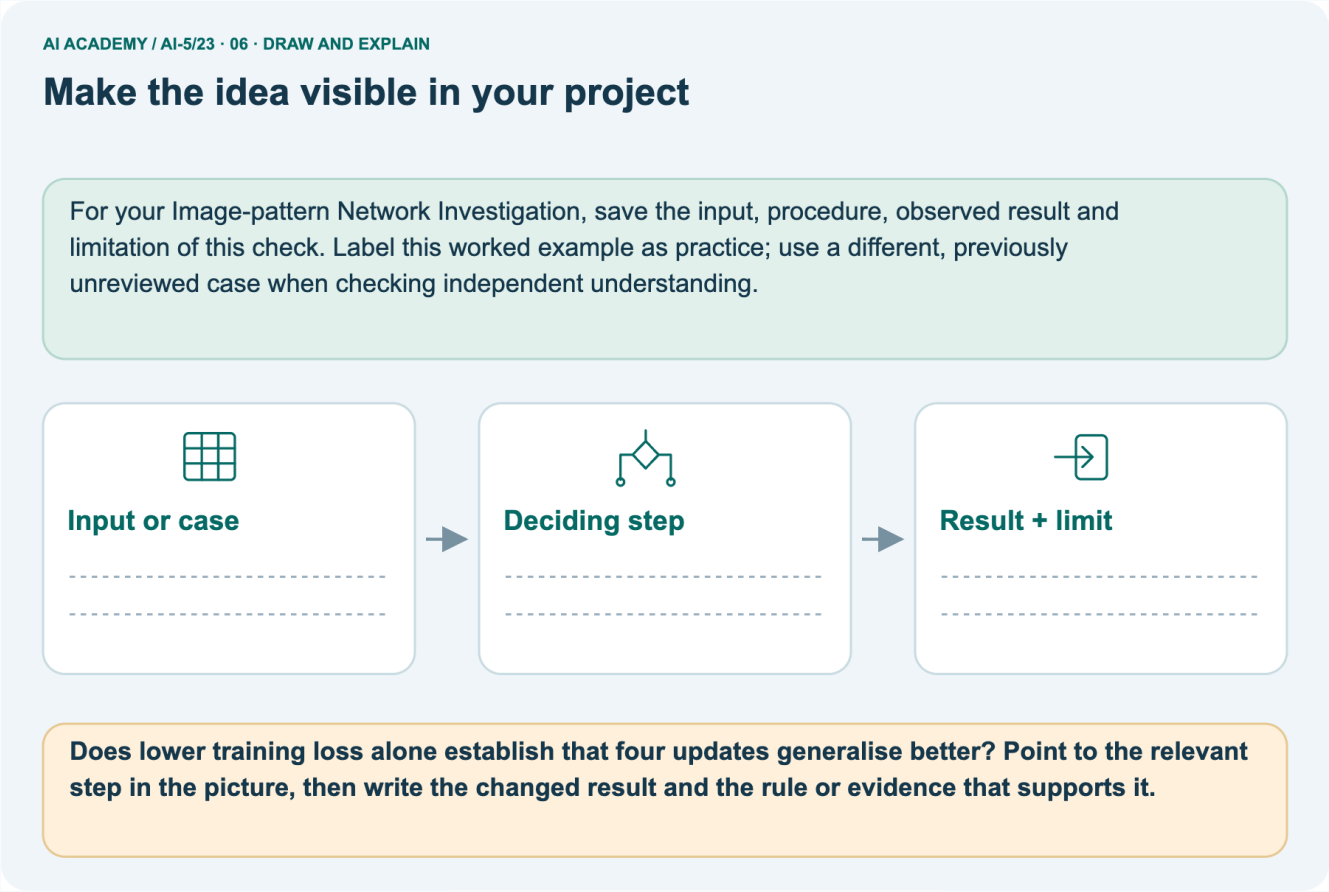
Limit: This scalar brightness model is a transparent toy image task, not a claim that a full image network was run.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This scalar brightness model is a transparent toy image task, not a claim that a full image network was run.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does lower **training** loss alone establish that four updates generalise better? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No. Record the extra budget and score suitable validation **evidence**; use an untouched final evaluation after selecting a schedule.

Changed-case check: Next a values 0.875 and 0.9375; **training** loss falls to 0.015625. Each update halves the distance to the **training** optimum 1 in this toy system.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Training a small image model: the case

02. Follow the deciding steps

03. Compare the saved checkpoints

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.